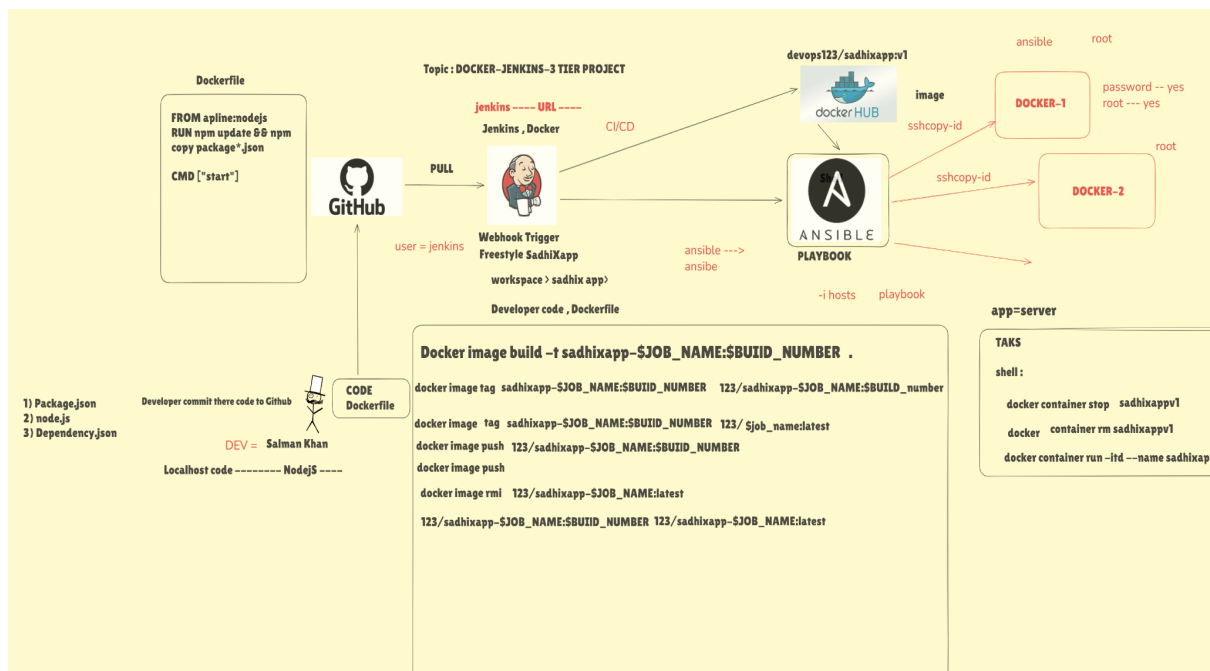


Docker + Jenkins + Ansible 3-Tier Deployment Pipeline

This document explains the complete CI/CD pipeline using Docker, Jenkins, and Ansible. The process includes code commit, image build, Docker Hub push, and automated deployment on multiple application servers using Ansible with SSH key authentication.

Architecture Overview



The pipeline consists of: 1. Developer commits code + Dockerfile to GitHub. 2. Jenkins pulls the repository using a webhook trigger. 3. Jenkins builds the Docker image using JOB_NAME and BUILD_NUMBER. 4. Jenkins tags and pushes images to Docker Hub. 5. Jenkins triggers Ansible. 6. Ansible connects to multiple Docker servers using SSH keys. 7. Ansible pulls latest image and restarts container on all servers.

1. SSH Key Setup

Generate SSH Key:
ssh-keygen -t rsa -b 4096

Copy key to servers:
ssh-copy-id root@DOCKER_SERVER_1
ssh-copy-id root@DOCKER_SERVER_2

2. Ansible Inventory File

```
/mnt/hosts Example:
[appservers]
54.172.171.132
54.210.55.200
```

3. Ansible Playbook

```
---
- name: Deploy sadhixapp container
  hosts: all
  become: yes

tasks:
- name: Stop existing container
  shell: docker container stop sadhixappv1 || true

- name: Remove existing container
  shell: docker container rm -f sadhixappv1 || true

- name: Remove old latest image
  shell: docker image rm maybtdevops/sadhixapp-project:latest || true

- name: Run new container
  shell: docker run -itd --name sadhixappv1 -p 80:3000 maybtdevops/sadhixapp-project:latest
```

4. Jenkins Build Script

```
cd /var/lib/jenkins/workspace/project/sadhixapp-project

docker build -t sadhixapp-${JOB_NAME}:${BUILD_NUMBER} .

docker tag sadhixapp-${JOB_NAME}:${BUILD_NUMBER}
maybtdevops/sadhixapp-${JOB_NAME}:${BUILD_NUMBER}
docker tag sadhixapp-${JOB_NAME}:${BUILD_NUMBER}
maybtdevops/sadhixapp-${JOB_NAME}:latest

docker push maybtdevops/sadhixapp-${JOB_NAME}:${BUILD_NUMBER}
docker push maybtdevops/sadhixapp-${JOB_NAME}:latest

docker rmi maybtdevops/sadhixapp-${JOB_NAME}:${BUILD_NUMBER}
docker rmi maybtdevops/sadhixapp-${JOB_NAME}:latest
```

This completes the full CI/CD workflow for Node.js + Docker + Jenkins + Ansible automation.